

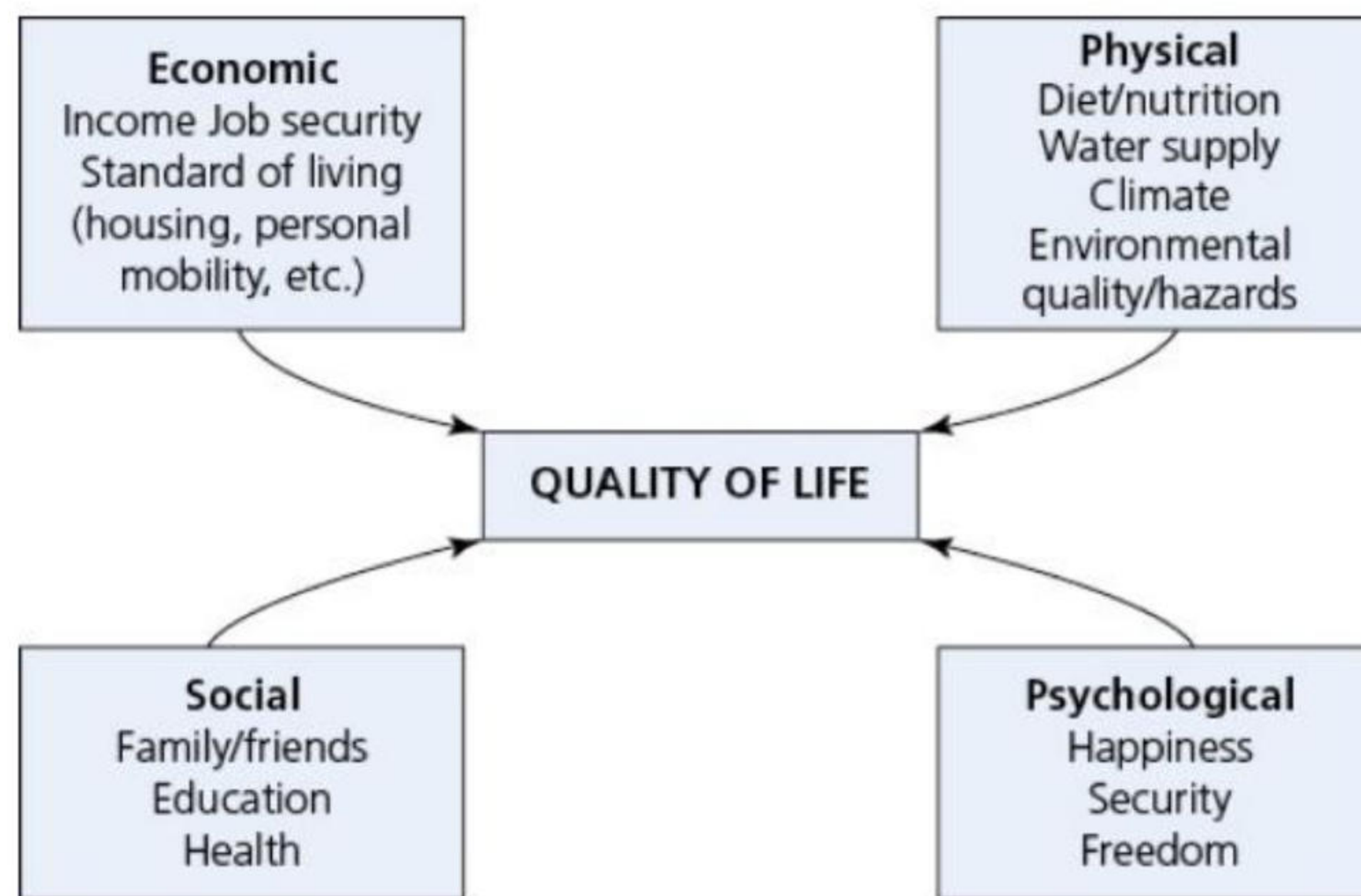
8.1 Measuring development

This chapter will explain:

- ★ indicators of development
- ★ the Human Development Index: a broader measure of development.

Indicators of development

Development, or improvement in the quality of life, is a wide-ranging concept. It includes increasing wealth, but also involves other important aspects of our lives ([Figure 8.1](#)). Many people would consider a good health service to be more important than wealth. People who live in countries that are not democracies, where freedom of speech cannot be taken for granted, often envy those who do live in democratic countries, though many African countries look to China as a model of success.



▲ **Figure 8.1** Factors comprising the quality of life

Development occurs when there are improvements to the individual factors making up the quality of life and standard of living ([Figure 8.2](#)). The more factors that are improved in terms of people's wellbeing, the greater the rate of overall progress. For example, development occurs in a low-income country when:

- » local food supply improves due to investment in farm machinery and fertilisers
- » the electricity grid extends outwards from the main urban areas to rural areas
- » levels of literacy improve throughout the country
- » a new road or railway improves the accessibility of a remote province.



▲ **Figure 8.2** An open-pit toilet in the Gobi Desert: in some parts of the world sanitation can be very basic

Thus, development is a process of change for the better. However, in some cases it can proceed in a way that adversely affects the most impoverished in a society. Development is not always a continuous process as it can be interrupted by major national and global events such as war, famine, disease and economic recession. The Covid-19 pandemic had a massive negative impact on development, and set back progress in LICs by many years.

The **United Nations Development Programme (UNDP)** is the UN's development agency working to eradicate poverty. It plays a crucial role in helping countries achieve the Sustainable Development Goals. The UNDP has highlighted a number of important issues concerning development:

- » Development is about creating opportunities and giving people the freedom to make choices.
- » Sustainability is vital for long-term development.
- » Development must balance social, economic and environmental sustainability.

[Table 8.1](#) looks at four categories of development, giving examples of how countries can improve these aspects of the quality of life of their citizens. It is not an exclusive list and you may well be able to think of other examples within each category.

▼ **Table 8.1** Categories of development and examples

Category	Examples
Social	● Achieving higher levels of literacy ● Reducing infant and maternal mortality ● Improving life expectancy ● Reducing the gender gap ● Improving the quality of housing and other aspects of soft infrastructure
Economic	● Raising productivity ● Increasing incomes for all levels of society ● Improving hard infrastructure (roads, railways, airports) ● Achieving a better balance between economic sectors ● Increasing trade with other countries ● Raising per capita energy consumption ● Increasing level of car ownership
Environmental	● Reducing CO₂ and other greenhouse gas emissions: cutting lower atmosphere air pollutants ● Improving water quality in rivers and other water bodies ● Tackling land pollution and other issues such as deforestation ● Conserving biodiversity
Political	● Improving democratic institutions ● Implementing measures to reduce corruption ● Keeping to new international human rights obligations

Economic indicators of development

There are three widely used economic indicators of development:

- » Gross domestic product (GDP) is the total value of goods and services produced within a country's borders, regardless of the nationality of those involved.
- » Gross national product (GNP) focuses on the income generated by a country's residents, regardless of location.
- » Gross national income (GNI) provides the most comprehensive picture by incorporating both GDP and GNP.

The World Bank, the Human Development Index and the annual World Population Data Sheet published by the Population Reference Bureau all use GNI to illustrate the development differences between countries. For all three of these economic indicators, both absolute and relative data is available. Relative data, for example **GNI per capita**, tends to be the most widely used. GNI per capita (see [Table 8.2](#)) is calculated by dividing the total GNI of a country for a given year by the number of people in that country. Per capita data allows comparisons to be made between countries that have big differences in their population size. For example, the total GNI of China is greater than that of France, but GNI per capita is much higher in France.

However, such data do not take into account the differences in the cost of living between countries. For example, a dollar buys much more in Bangladesh than it does in the USA. To account for this, the GNI per capita at **purchasing power parity (PPP)** can be calculated.

While GNI per capita is a useful measure, it does not tell us anything about:

- » wealth distribution within a country – in some countries the gap between highest and lowest incomes is much greater than in others
- » the ways in which governments invest the money at their disposal – a number of factors can affect the efficiency with which governments operate, with a high level of **corruption** being a significant problem in some countries.

Economic growth (GNI growth) refers to an increase in real GNI (percentage in a year). ‘Real’ means after inflation is taken into account. Countries that experience high rates of inflation often struggle to cope with rising prices as average incomes lag behind. A high rate of inflation can adversely affect economic confidence within a country itself and also impact investment coming into the country from abroad.

Activities

- 1 What is development?
- 2 Give two examples of development in a low-income country.
- 3 The UNDP stresses the importance of a balance between which three types of development?
- 4 What is the reason for calculating GNI data at purchasing power parity?

▼ **Table 8.2** Key development indicators by world region

Region	Life expectancy at birth (years)	Infant mortality rate (per 1000)	GNI per capita (US\$)	Per capita kilocalorie supply from all foods per day (kcal)
World	72	29	25,100	2959
Africa	63	44	5645	2573
North America	77	5	75,470	3587
Latin America/Caribbean	74	13	18,412	3108
Asia	74	24	16,349	2927
Europe	78	4	47,960	3456
Oceania	79	15	43,904	3087

Other indicators of development

Many individual indicators other than those above, which are clearly economic, have been used to highlight contrasts in development. Some of the most widely used are the literacy rate, life expectancy at birth, infant mortality, calorie intake and number of people per doctor. Development varies not only between countries – it can also vary significantly within countries. For instance, the number of people per doctor is likely to be higher in rural areas. Most of the measures that can be used to study the contrasts between countries can also be used to look at regional variations within countries.

Literacy

Education is the key to socioeconomic development. It can be defined as the process of acquiring knowledge, understanding and skills. Education has always been regarded as a very important indicator of development and it has figured prominently in aggregate/composite measures of development. Adult literacy (the ability to read and write) is one of the main ways in which differences in educational standards between countries can be shown. In 2020, the global **adult literacy rate** was estimated at 86 per cent. For developed nations it is at least 96 per cent, but for LICs the average adult literacy rate is only 65 per cent. A low adult literacy rate is a great obstacle to development. Poverty and illiteracy go hand in hand. Many of the countries with the lowest literacy rates are in Sub-Saharan Africa.

Of the roughly 780 million adults worldwide who cannot read and write, nearly two-thirds are female. Improving female literacy is essential because so many aspects of development depend on it. For example, there is a very strong relationship between levels of female literacy and infant mortality rates. People who are literate are much more able to access medical and other information that will help them achieve a higher quality of life compared with those who are illiterate.

Life expectancy at birth

Life expectancy at birth is to a large extent the end result of all the factors contributing to the quality of life in a country. The main influences on life expectancy are:

- » the incidence of disease, e.g. malaria ([Figure 8.3](#))
- » physical environmental conditions, e.g. very low rainfall
- » human environmental conditions, e.g. pollution
- » personal lifestyle, e.g. smoking.

Rates of life expectancy in HICs and LICs have converged significantly during the last 50 years, in spite of a

widening wealth gap. [Table 8.2](#) shows that the lowest life expectancy by world region is in Africa, at 63 years. However, in 2014, Africa’s life expectancy was 59 years. It is encouraging that some aspects of development can improve quite quickly!

Infant mortality

The **infant mortality rate** is regarded as one of the most sensitive indicators of socioeconomic progress. It is an important measure of health inequality between and within countries. There are huge differences in infant mortality around the world, despite the wide availability of public health knowledge. The main causal factors of such disparities are:

- » contrasts in material resources
- » differences in the efficiency of social institutions and health systems.



▲ **Figure 8.3** A public health poster in a park in Buenos Aires, Argentina – disease control is an important aspect of development

Fortunately, infant mortality rates have fallen sharply in many MICs and LICs over the last 30 years. Nevertheless, [Table 8.2](#) shows that the infant mortality rate in Africa is 11 times that in Europe. Sixteen countries had an infant mortality rate of 50/1000 or more in 2023 – this means one in 20 infants dying before their first birthday. All 16 of those countries are in Sub-Saharan Africa, with the highest levels of infant mortality being in Sierra Leone (75/1000) and Lesotho (69/1000). [Table 8.3](#) shows the extent to which infant mortality varies by region in Africa. Some countries have registered very large reductions in infant mortality in recent decades, while others have seen reductions at a lower rate.

▼ **Table 8.3** Infant mortality rate by region in Africa, 2023

Region	Number of countries above 50/1000	Average infant mortality rate
Northern Africa	0	23
Western Africa	8	54
Eastern Africa	3	39
Middle Africa	4	51
Southern Africa	1	27

Infant mortality generally compares well with many other indicators of development, which is a good indication of its value as a measure of development.

Daily calorie intake

Daily calorie intake is viewed as an important measure because people need energy, which is provided by calories, for their bodies to function properly. While the average figures by world region are useful in showing general contrasts around the world, it is important to remember that the countries with the lowest average calorie intake are those that usually suffer the greatest degree of food insecurity. During times of drought and famine, daily calorie intake may only be a fraction of the 'average' figure. A general daily guideline for weight maintenance is 2000 calories for women and 2500 for men. However, excessive calorie intake can result in obesity and associated metabolic diseases.

Doctors per thousand people

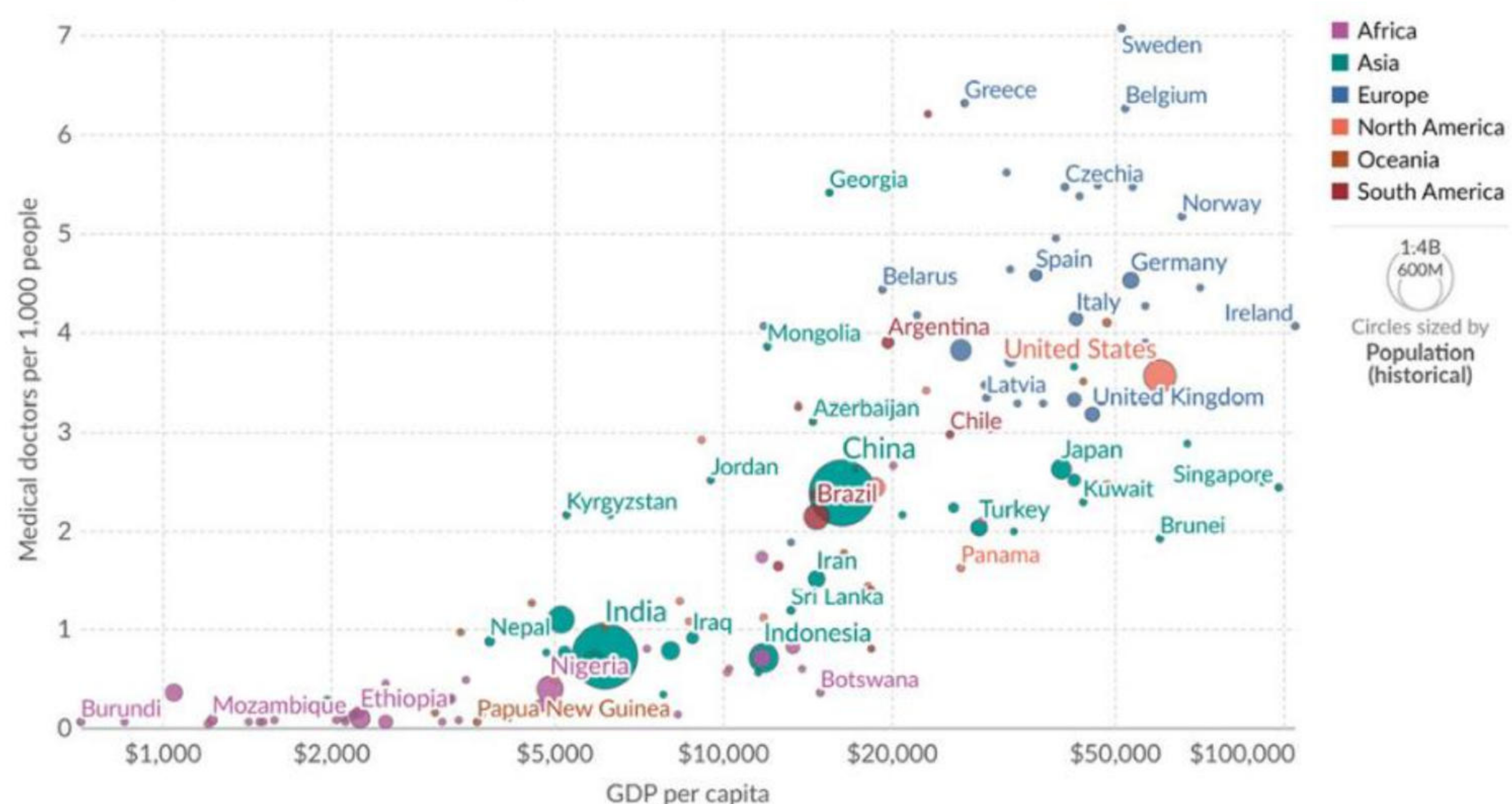
The number of doctors per unit of population is an indication of the financial resources available to a country for the training and recruitment of doctors. Access to well-qualified medical professionals impacts directly on the wellbeing of patients. [Figure 8.4](#) is a scatter graph that shows the relationship between medical doctors per thousand people and GDP per capita for a wide range of countries. The graph shows that most, but not all, HICs have more than three doctors per thousand people, while LICs generally have less than one. In fact many LICs are in the situation that they are nowhere near one doctor per thousand people – look how many LICs are very close to the baseline (the horizontal axis) on the diagram. Overall, the scatter graph shows that there is a strong correlation (relationship) between doctors per thousand people and GDP per capita.

Interesting note

Corruption is a major obstacle to economic and social development around the world. The Corruption Perceptions Index has been published annually since 1995 by Transparency International. It ranks the countries of the world for perceived levels of public sector corruption.

Medical doctors per 1,000 people vs. GDP per capita, 2021

Medical doctors include generalist physicians and specialist medical practitioners. GDP per capita is adjusted for inflation and differences in the cost of living between countries.



▲ **Figure 8.4** The relationship between doctors per 1000 people and GDP per capita

The relationship between different indicators of development

All indicators of human development have merits and limitations. The indicators discussed above have been widely used because their merits are generally considered to be greater than their limitations. Therefore, it is reasonable to assume that you might expect there to be a strong (but not perfect) relationship between any pair of these indicators, such as that illustrated by [Figure 8.4](#).

Activities

- 1 Define adult literacy.

- 2 Which world regions have the highest and lowest infant mortality rates?
- 3 List the four main influences on life expectancy.
- 4 Using the data from [Table 8.2 \(page 180\)](#), draw a scatter graph to show the relationship between life expectancy at birth and infant mortality rate. Insert a line of best fit and comment on the relationship between the two variables.

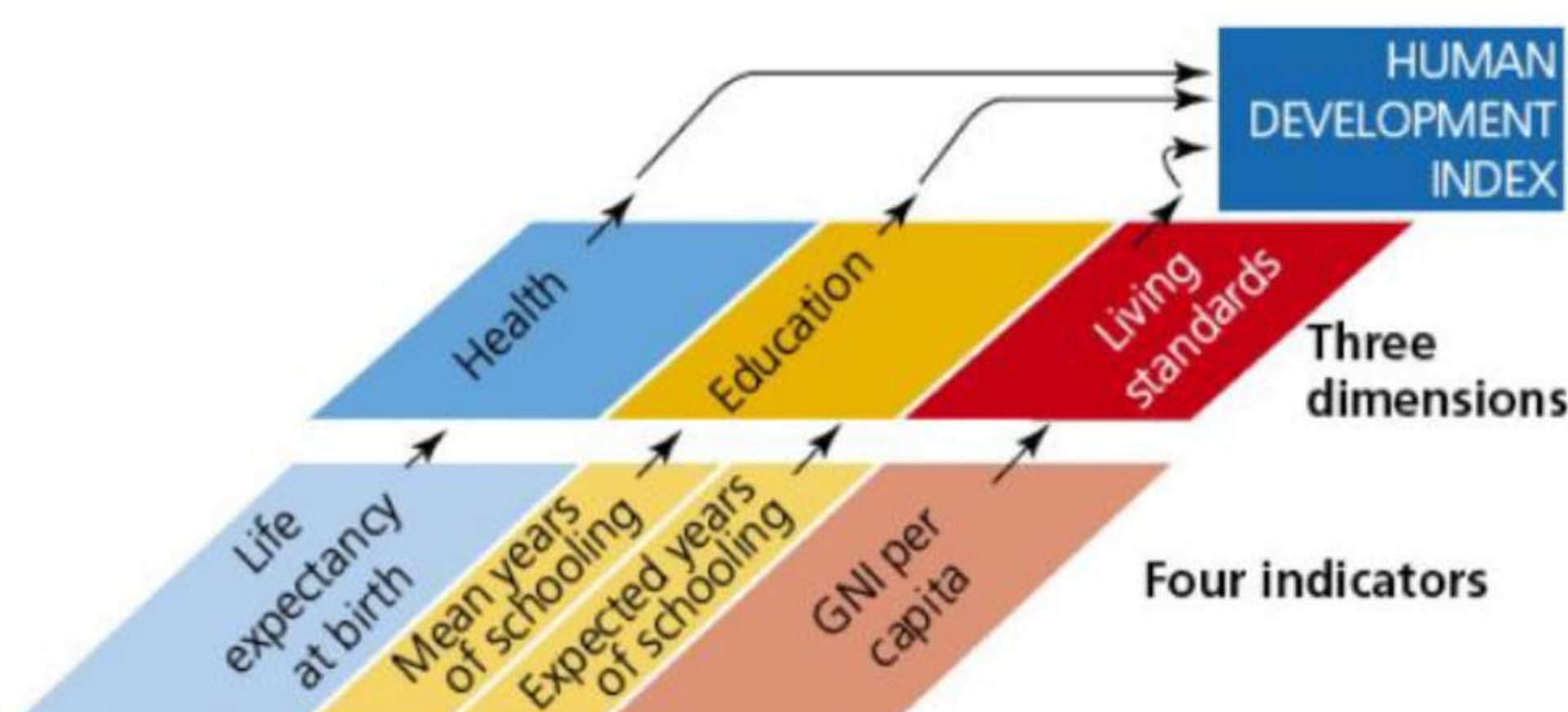
The Human Development Index: a broader measure of development

The **Human Development Index (HDI)** was devised by the United Nations in 1990. It is a composite index, combining a number of key development indicators. Its aim was to extend the concept of development beyond wealth alone. The United Nations Development Programme (UNDP), which is responsible for producing the HDI, has stated that 'The HDI is a summary measure of average achievement in key dimensions of human development – a long and healthy life, being knowledgeable and having a decent standard of living'. Canada ([Figure 8.5](#)) has a very high level of human development by these measures, but the UNDP acknowledges that the HDI includes only part of what human development entails. For example, it does not include measures of inequality, poverty, human security or empowerment.



▲ **Figure 8.5** The Waterfront, Vancouver: Canada has a very high level of human development

The HDI has changed slightly in composition over the years. The current index combines four indicators of development ([Figure 8.6](#)):

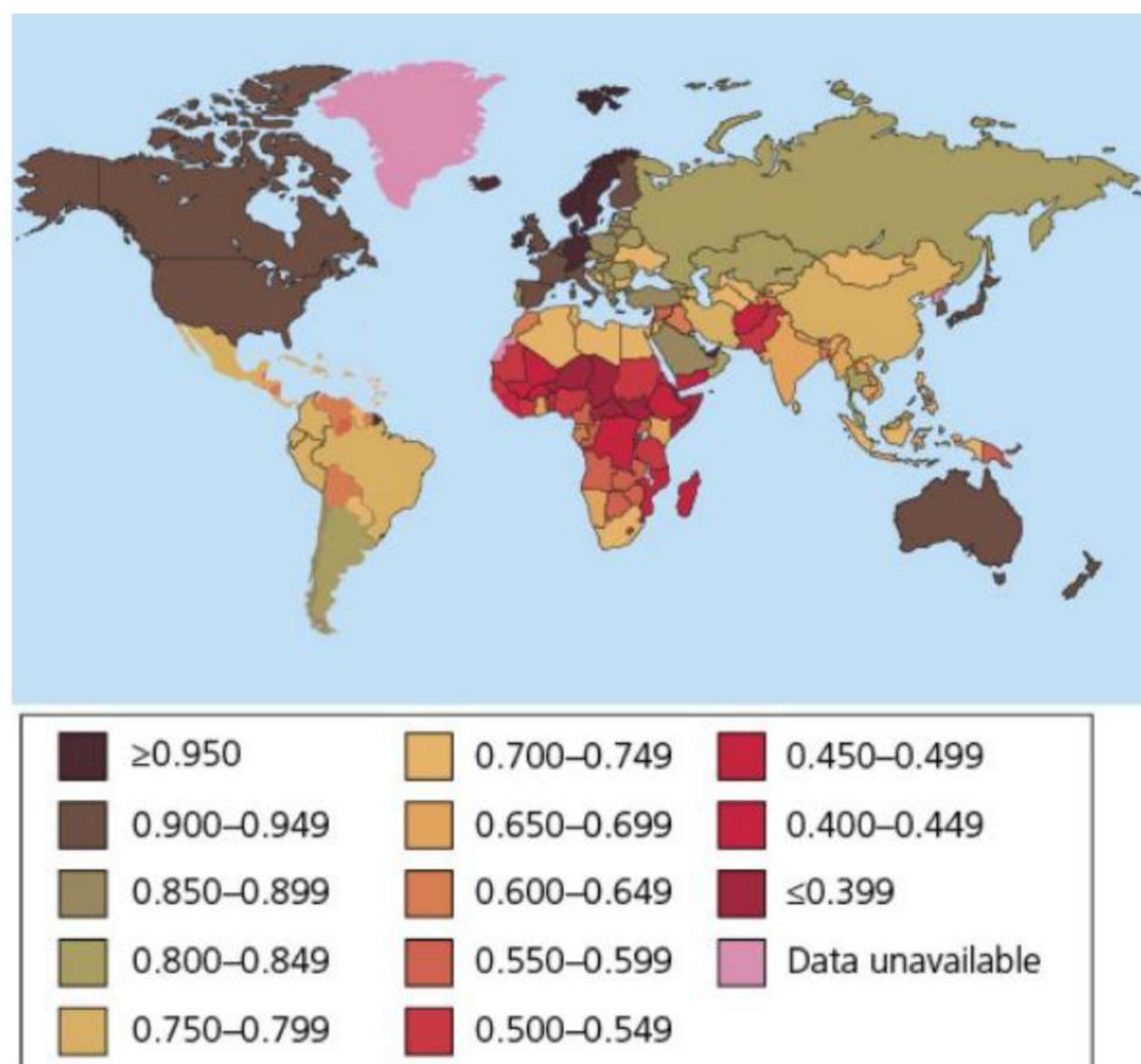


▲ **Figure 8.6** The components of the Human Development Index

- » Life expectancy at birth
- » Mean years of schooling for adults aged 25 years
- » Expected years of schooling for children of school-entering age
- » GNI per capita (\$PPP).

The actual figures for each of these measures are converted into an index, which has a maximum value of 1.0 in each case. The index figures are then combined and averaged to give an overall HDI value. This also has a maximum value of 1.0 ([Figure 8.7](#)). The United Nations publishes an updated Human Development Report every year, which uses the HDI to rank all the countries of the world on their level of development. The HDI divides the countries of the world into four groups. The 2024 report, based on 2022 data, divided the countries of the world as follows:

- » Very high HDI: The countries ranked 1–69
- » High HDI: The countries ranked 70–115
- » Medium HDI: The countries ranked 116–159
- » Low HDI: The countries ranked 160–193.



▲ **Figure 8.7** The Human Development Index scores in increments of 0.050, from the 2024 report, based on 2022 data

[Table 8.4](#) shows examples of countries in each group in the 2024 report. Some countries have progressed from one group to another in a relatively short period of time.

▼ **Table 8.4** Examples of countries in each group in the 2024 report

Level of development	Example countries
Very high human development	Norway, Germany, Canada, UK, Japan, Australia
High human development	Cuba, Mexico, Brazil, China, South Africa, Indonesia
Medium human development	India, Morocco, Ghana, Pakistan, Philippines, Bangladesh
Low human development	Nigeria, Tanzania, Afghanistan, DR Congo, Yemen, Mozambique

Activities

- 1 Which indicators of development are combined to form the Human Development Index?
- 2 Look at [Figure 8.7](#) and briefly describe the distribution of countries according to the Human Development Index.